Lecture 1.

Inexact Nesterov's accelerated proximal gradient method

Yangyang Xu

xuy21@rpi.edu

Motivated by applications where only approximate gradients are available, we study an inexact variant of Nesterov's accelerated proximal gradient (APG) method given in [1]. Though inexact APG has been explored in literature, e.g., [2, 3], Nesterov's APG achieves accelerated convergence for both convex and strongly convex problems in a unified framework.

1.1 Problem Setting

We consider the following structured problem

$$x^* = \underset{x \in \mathbb{R}^n}{\arg\min}\, \phi(x) := f(x) + \Psi(x), \tag{1.1}$$

where f is convex and L_f-smooth on $\mathbb{R}^n$, and $\Psi(x)$ is closed and μ-strongly convex with $\mu > 0$. We assume that for any $\eta > 0$, the proximal mapping of $\eta\Psi$ can be computed exactly, i.e.,

$$\mathbf{prox}_{\eta\Psi}(u) := \arg\min_x \frac{1}{2}\|x-u\|^2 + \eta\Psi(x) \tag{1.2}$$

can be computed for each $u \in \mathbb{R}^n$. However, an exact gradient of f is not available; instead its approximate gradient at any x, denoted as $\tilde{\nabla} f(x)$, can be accessed. For each $x \in \mathbb{R}^n$, we denote

$$e(x) = \tilde{\nabla} f(x) - \nabla f(x) \tag{1.3}$$

as the error of the approximate gradient.

1.2 Inexact APG with line search

With the inexact gradient oracle, we give the inexact variant of Nesterov's APG in Algorithm 1 for solving (1.1). Notice that we assume to know the strong convexity constant $\mu > 0$. Nevertheless, by performing line search, we do not need to know the smoothness constant L_f. Except using inexact gradients, other settings, including the line search strategy and choice of coefficients a_k and A_k, are the same as those in [1]. A key difference lies at the step to obtain a near-stationary solution.

By the definition of y_k, x_{k+1}, τ_k and A_{k+1} in Algorithm 1, we have

$$\max\left\{\|e(y_k)\|, \|e(x_{k+1})\|\right\} \le \tau_k = \min\left\{\frac{\varepsilon}{8}, \frac{\mu\varepsilon}{(k+1)^2 \cdot \max\{1, A_{k+1}\}}\right\}, \forall\, k \ge 0. \tag{1.4}$$

Remark 1. Notice that if there is no gradient error, Algorithm 1 reduces to the exact version of Nesterov's APG.

1.3 Convergence Analysis

We follow the analysis in [1] but extend a few of its key results to accommodate the usage of inexact gradients. First, we show the while-loop for line search must stop within a finite number of steps. Based on the next lemma, we assume $L_{\min} \le L_f$ and $L_0 \le L_f$ without loss of generality.

Algorithm 1: Inexact accelerated proximal gradient method with line search for (1.1)

Input: $\varepsilon > 0$, $x_0 \in \mathrm{dom}(\Psi)$, $\gamma_u > 1$, $\gamma_d > 1$, and $L_{\min} > 0$.
Set: $A_0 = 0$, $\psi_0(x) = \frac{1}{2}\|x - x_0\|^2$, $L_0 \geq L_{\min}$, and $v_0 = x_0$.
for $k = 0, 1, \ldots$ **do**
  Let $L = L_k$. Set $\texttt{FLAG} =$ false.
  while $\texttt{FLAG} ==$ *false* **do**
    Find $a > 0$ such that $\frac{a^2}{A_k + a} = \frac{2(1+\mu A_k)}{L}$; let $y = \frac{A_k x_k + a v_k}{A_k + a}$.
    Let $\tau = \min\left\{\frac{\varepsilon}{8}, \frac{\mu\varepsilon}{(k+1)^2 \cdot \max\{1, A_k + a\}}\right\}$; obtain $\tilde{\nabla} f(y)$ satisfying $\|e(y)\| \leq \tau$.
    Set $z = \mathbf{prox}_{\Psi/L}\left(y - \frac{1}{L}\tilde{\nabla} f(y)\right)$ and let $\xi = \tilde{\nabla} f(z) - \tilde{\nabla} f(y) - L(z - y)$.
    Obtain $\tilde{\nabla} f(z)$ satisfying $\|e(z)\| \leq \tau$.
    if $\langle \xi, y - z\rangle \geq \frac{1}{L}\|\xi\|^2 - 6\tau\|y - z\| - \frac{4\tau^2}{L}$ **then**
      $\texttt{FLAG} =$ true.
    else
      $L = L \cdot \gamma_u$.
  Let $y_k = y$, $x_{k+1} = z$, $M_k = L$, $L_{k+1} = \max\{L_{\min}, L/\gamma_d\}$, $a_{k+1} = a$, $A_{k+1} = A_k + a$, and $\tau_k = \tau$.
  Set $v_{k+1} = \arg\min_x \psi_{k+1}(x) := \psi_k(x) + a_{k+1}\left(f(x_{k+1}) + \langle \tilde{\nabla} f(x_{k+1}), x - x_{k+1}\rangle + \Psi(x)\right)$.
  if $\|\tilde{\nabla} f(z) - \tilde{\nabla} f(y) - L(z - y)\| \leq \frac{\varepsilon}{2}$ **then**
    Return z and Stop.

Lemma 1. For $\xi = \tilde{\nabla} f(z) - \tilde{\nabla} f(y) - L(z - y)$, when $L \geq L_f$, it must hold that

$$\langle \xi, y - z\rangle \geq \frac{1}{L}\|\xi\|^2 - 3\|e(y) - e(z)\| \cdot \|y - z\| - \frac{1}{L}\|e(y) - e(z)\|^2.$$

Consequently, the while-loop in Algorithm 1 must exit after at most $\left\lceil \log_{\gamma_u} \frac{L_f}{L_{\min}} \right\rceil + 1$ steps.

Proof. By $\xi = \tilde{\nabla} f(z) - \tilde{\nabla} f(y) - L(z - y)$, it holds

$$\begin{aligned}
&\langle \xi,\, y - z\rangle \\
&= L\|y - z\|^2 - \left\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(z),\, y - z\right\rangle \\
&= \frac{1}{L}\Big(\|\xi\|^2 + 2L\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(z),\, y - z\rangle - \|\tilde{\nabla} f(y) - \tilde{\nabla} f(z)\|^2\Big) - \left\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(z),\, y - z\right\rangle \\
&= \frac{1}{L}\|\xi\|^2 + \left\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(z),\, y - z\right\rangle - \frac{1}{L}\|\tilde{\nabla} f(y) - \tilde{\nabla} f(z)\|^2.
\end{aligned}$$

Hence, using the definition of $e(\cdot)$ in (1.3), we have

$$\begin{aligned}
\langle \xi,\, y - z\rangle = {} & \frac{1}{L}\|\xi\|^2 + \langle \nabla f(y) - \nabla f(z),\, y - z\rangle - \frac{1}{L}\|\nabla f(y) - \nabla f(z)\|^2 \\
& + \langle e(y) - e(z),\, y - z\rangle - \frac{1}{L}\|e(y) - e(z)\|^2 - \frac{2}{L}\langle \nabla f(y) - \nabla f(z),\, e(y) - e(z)\rangle \\
\geq {} & \frac{1}{L}\|\xi\|^2 + \langle \nabla f(y) - \nabla f(z),\, y - z\rangle - \frac{1}{L}\|\nabla f(y) - \nabla f(z)\|^2 \qquad (1.5) \\
& - \left(1 + \tfrac{2L_f}{L}\right)\|e(y) - e(z)\| \cdot \|y - z\| - \tfrac{1}{L}\|e(y) - e(z)\|^2,
\end{aligned}$$

where the inequality follows from $\langle \nabla f(y) - \nabla f(z),\, e(y) - e(z)\rangle \leq L_f\|e(y) - e(z)\| \cdot \|y - z\|$ by the L_f-smoothness of f and the Cauchy-Schwarz inequality.

Now when $L \geq L_f$, we have $1 + \frac{2L_f}{L} \leq 3$ and from the convexity and L_f-smoothness of f that

$$\langle \nabla f(y) - \nabla f(z),\, y-z\rangle - \frac{1}{L}\|\nabla f(y)-\nabla f(z)\|^2 \geq \langle \nabla f(y)-\nabla f(z),\, y-z\rangle - \frac{1}{L_f}\|\nabla f(y)-\nabla f(z)\|^2 \geq 0.$$

Hence, when $L \geq L_f$, the inequality in (1.5) implies the desired result. $\square$

The next lemma extends one key relation in [1] to the inexact case.

Lemma 2. Let $\delta_0 = 0$ and define $\{\delta_k\}_{k\geq 1}$ by

$$\delta_{k+1} = \delta_k + A_{k+1}\,\varepsilon_k + A_k\tau_k\|x_k - x_{k+1}\|,\ \forall\, k \geq 0, \tag{1.6}$$

with

$$\varepsilon_k = 6\tau_k\|y_k - x_{k+1}\| + \frac{4\tau_k^2}{M_k},\ \forall\, k \geq 0. \tag{1.7}$$

Also, let $\psi_k^* = \min_x \psi_k(x)$ for each $k \geq 0$. Then it holds that

$$A_k\phi(x_k) \leq \psi_k^* + \delta_k,\ \forall\, k \geq 0. \tag{$\mathcal{R}_k^1$}$$

Proof. Since $A_0 = 0$, $\delta_0 = 0$, and $\psi_0^* = 0$, the inequality in ($\mathcal{R}_k^1$) clearly holds for $k = 0$. Now suppose it holds at k. We show it also holds for $k+1$ as follows.

By the $(1+\mu A_k)$–strong convexity of ψ_k, we have

$$\psi_k(x) \geq \psi_k^* + \frac{1+\mu A_k}{2}\|x - v_k\|^2 \geq A_k\phi(x_k) - \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2. \tag{1.8}$$

Hence, for any $g_{k+1} \in \partial\Psi(x_{k+1})$, it holds

$$\begin{aligned}
\psi_{k+1}^* &= \min_x \Big\{\psi_k(x) + a_{k+1}\big[f(x_{k+1}) + \langle\tilde\nabla f(x_{k+1}), x - x_{k+1}\rangle + \Psi(x)\big]\Big\}\\
&\overset{(1.8)}{\geq} \min_x \Big\{A_k\phi(x_k) - \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2\\
&\qquad\qquad + a_{k+1}\big[\phi(x_{k+1}) + \langle\tilde\nabla f(x_{k+1}) + g_{k+1}, x - x_{k+1}\rangle\big]\Big\}\\
&\geq \min_x \Big\{A_k\big[\phi(x_{k+1}) + \langle\nabla f(x_{k+1}) + g_{k+1}, x_k - x_{k+1}\rangle\big] - \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2\\
&\qquad\qquad + a_{k+1}\big[\phi(x_{k+1}) + \langle\tilde\nabla f(x_{k+1}) + g_{k+1}, x - x_{k+1}\rangle\big]\Big\}\\
&= \min_x \Big\{A_{k+1}\phi(x_{k+1}) + \langle\nabla f(x_{k+1}) + g_{k+1}, A_k(x_k - x_{k+1})\rangle - \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2\\
&\qquad\qquad + a_{k+1}\langle\tilde\nabla f(x_{k+1}) + g_{k+1}, x - x_{k+1}\rangle\Big\}\\
&= \min_x \Big\{A_{k+1}\phi(x_{k+1}) - \langle e(x_{k+1}), A_k(x_k - x_{k+1})\rangle - \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2\\
&\qquad + \langle\tilde\nabla f(x_{k+1}) + g_{k+1}, A_k(x_k - x_{k+1})\rangle + a_{k+1}\langle\tilde\nabla f(x_{k+1}) + g_{k+1}, x - x_{k+1}\rangle\Big\}\\
&= \min_x \Big\{A_{k+1}\phi(x_{k+1}) - \langle e(x_{k+1}), A_k(x_k - x_{k+1})\rangle - \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2\\
&\qquad + A_{k+1}\langle\tilde\nabla f(x_{k+1}) + g_{k+1}, y_k - x_{k+1}\rangle + a_{k+1}\langle\tilde\nabla f(x_{k+1}) + g_{k+1}, x - v_k\rangle\Big\},
\end{aligned} \tag{1.9}$$

where the second inequality follows from the convexity of ϕ, and the last equality holds by the relation $A_kx_k = A_{k+1}y_k - a_{k+1}v_k$.

Notice that the minimum in (1.9) is achieved at $v_k - \frac{a_{k+1}}{1+\mu A_k}(\tilde\nabla f(x_{k+1}) + g_{k+1})$. Thus,

$$\psi_{k+1}^* \geq A_{k+1}\phi(x_{k+1}) - \langle e(x_{k+1}), A_k(x_k - x_{k+1})\rangle + A_{k+1}\langle\tilde\nabla f(x_{k+1}) + g_{k+1}, y_k - x_{k+1}\rangle - \delta_k$$

$$- \frac{a_{k+1}^2}{2(1+\mu A_k)} \|\tilde{\nabla} f(x_{k+1}) + g_{k+1}\|^2. \tag{1.10}$$

Now choose $g_{k+1} = -\tilde{\nabla} f(y_k) - M_k(x_{k+1} - y_k) \in \partial \Psi(x_{k+1})$ in the above inequality and use the exit condition for the first while-loop in Algorithm 1, i.e., $\langle \tilde{\nabla} f(x_{k+1}) + g_{k+1}, y_k - x_{k+1}\rangle \geq \frac{1}{M_k}\|\tilde{\nabla} f(x_{k+1}) + g_{k+1}\|^2 - \varepsilon_k$, to obtain from (1.10) and the Cauchy-Schwarz inequality that

$$\begin{aligned}
\psi_{k+1}^* &\geq A_{k+1}\phi(x_{k+1}) - A_k\|e(x_{k+1})\|\|x_k - x_{k+1}\| - A_{k+1}\varepsilon_k - \delta_k \\
&\quad + \left(\frac{A_{k+1}}{M_k} - \frac{a_{k+1}^2}{2(1+\mu A_k)}\right)\|\tilde{\nabla} f(x_{k+1}) + g_{k+1}\|^2 \\
&\geq A_{k+1}\phi(x_{k+1}) - \delta_{k+1},
\end{aligned}$$

where the last inequality follows from the definition of δ_{k+1}, $\|e(x_{k+1})\| \leq \tau_k$, and the choice of a_{k+1}. Hence, we complete the induction and finish the proof. □

The lemma below extends another key relation in [1] to the inexact case.

Lemma 3. Let $B_0(x) \equiv 0, \forall\, x \in \mathbb{R}^n$ and define the sequence of functions $\{B_k(\cdot)\}_{k\geq 1}$ by

$$B_{k+1}(x) = B_k(x) + a_{k+1}\langle e(x_{k+1}), x - x_{k+1}\rangle, \ \forall\, k \geq 0. \tag{1.11}$$

Then it holds

$$\psi_k(x) \leq A_k\phi(x) + \frac{1}{2}\|x - x_0\|^2 + B_k(x), \forall\, x \in \mathrm{dom}(\Psi), \ \forall\, k \geq 0. \tag{$\mathcal{R}_k^2$}$$

Proof. By the definition of $B_0(\cdot)$, the choice of $\psi_0(\cdot)$, and $A_0 = 0$, the inequality in ($\mathcal{R}_k^2$) clearly holds for $k = 0$. Below we prove it by induction. Suppose it holds for some k. Then we have

$$\begin{aligned}
\psi_{k+1}(x) &= \psi_k(x) + a_{k+1}\Big[f(x_{k+1}) + \langle \tilde{\nabla} f(x_{k+1}),\, x - x_{k+1}\rangle + \Psi(x)\Big] \\
&\leq A_k\phi(x) + \frac{1}{2}\|x - x_0\|^2 + B_k(x) + a_{k+1}[f(x_{k+1}) + \langle \nabla f(x_{k+1}),\, x - x_{k+1}\rangle + \Psi(x)] \\
&\quad + a_{k+1}\langle e(x_{k+1}),\, x - x_{k+1}\rangle \\
&\leq A_k\phi(x) + \frac{1}{2}\|x - x_0\|^2 + B_k(x) + a_{k+1}[f(x) + \Psi(x)] + a_{k+1}\langle e(x_{k+1}),\, x - x_{k+1}\rangle \\
&= A_{k+1}\phi(x) + \frac{1}{2}\|x - x_0\|^2 + B_{k+1}(x),
\end{aligned}$$

where the first inequality follows from the induction assumption, and the second one is by the convexity of f. Hence, we finish the induction and complete the proof. □

With ($\mathcal{R}_k^1$) and ($\mathcal{R}_k^2$), we are ready to establish the convergence rate result.

Theorem 1 (Convergence Rate with Error Terms)**.** For each $k \geq 0$, it holds

$$\frac{1+\mu A_k}{2}\|x^* - v_k\|^2 \leq \frac{1}{2}\|x^* - x_0\|^2 + B_k(x^*) + \delta_k \tag{1.12}$$

and

$$\phi(x_k) - \phi(x^*) \leq \frac{1}{A_k}\left[\frac{1}{2}\|x^* - x_0\|^2 + B_k(x^*) + \delta_k\right]. \tag{1.13}$$

Proof. By ($\mathcal{R}_k^1$) and ($\mathcal{R}_k^2$), we have

$$A_k\phi(x_k) + \frac{1+\mu A_k}{2}\|x - v_k\|^2 \overset{(\mathcal{R}_k^1)}{\leq} \psi_k^* + \delta_k + \frac{1+\mu A_k}{2}\|x - v_k\|^2 \leq \psi_k(x) + \delta_k$$

$$\stackrel{(\mathcal{R}_k^2)}{\leq} A_k\phi(x) + \frac{1}{2}\|x - x_0\|^2 + B_k(x) + \delta_k,$$

where the second inequality follows from the $(1 + \mu A_k)$-strong convexity of ψ_k. Letting $x = x^*$ in the above inequality gives the desired results. □

The setting of a_k and A_k is the same as that in [1]. Hence, we directly have the following result from Lemma 8 in [1] for the lower bound of A_k.

Lemma 4. The coefficient sequence $\{A_k\}$ from Algorithm 1 satisfies the lower bound

$$A_k \geq \max\left\{\frac{k^2}{2\gamma_u L_f}, \frac{2}{\gamma_u L_f}\left(1 + \sqrt{\frac{\mu}{2\gamma_u L_f}}\right)^{2(k-1)}\right\}, \forall\, k \geq 1. \tag{1.14}$$

We have from (1.11) and the Cauchy-Schwarz inequality that

$$B_{k+1}(x^*) \leq B_k(x^*) + a_{k+1}\|e(x_{k+1})\| \cdot \|x^* - x_{k+1}\| \leq B_k(x^*) + \frac{\mu\varepsilon}{(k+1)^2}\|x^* - x_{k+1}\|$$

where the second inequality follows from (1.4) and $a_{k+1} \leq A_{k+1}$. Recursively applying the above inequality and noticing $B_0(\cdot) \equiv 0$ gives

$$B_{k+1}(x^*) \leq \mu\varepsilon\sum_{t=0}^{k}\frac{1}{(t+1)^2}\|x^* - x_{t+1}\|, \qquad \forall\, k \geq 0. \tag{1.15}$$

In addition, by the definition of δ_k in (1.6), it follows

$$\begin{aligned}\delta_{k+1} &= \delta_k + A_k\tau_k\|x_k - x_{k+1}\| + A_{k+1}\left(6\tau_k\|y_k - x_{k+1}\| + \frac{4\tau_k^2}{M_k}\right)\\ &\leq \delta_k + \frac{\mu\varepsilon}{(k+1)^2}\|x_k - x_{k+1}\| + \frac{6\mu\varepsilon}{(k+1)^2}\|y_k - x_{k+1}\| + \frac{4\mu^2\varepsilon^2}{M_k(k+1)^4},\end{aligned}$$

where we have used the value of τ_k in (1.4) to obtain the inequality. Hence, applying the above inequality recursively and noting $M_k \geq L_{\min}$, we have

$$\delta_{k+1} \leq \mu\varepsilon\sum_{t=0}^{k}\left(\frac{1}{(t+1)^2}\|x_t - x_{t+1}\| + \frac{6}{(t+1)^2}\|y_t - x_{t+1}\| + \frac{4\mu\varepsilon}{L_{\min}(t+1)^4}\right), \forall\, k \geq 0. \tag{1.16}$$

1.3.1 Bound on iterates

We show that all involved points in Algorithm 1 are bounded.

Lemma 5. For all $k \geq 0$, it holds that

$$\|y_k - x^*\| \leq R, \qquad \|v_k - x^*\| \leq R, \qquad \|x_k - x^*\| \leq R, \tag{1.17}$$

where

$$\begin{aligned}R = \max\Bigg\{&2\|x^* - x_0\| + \sqrt{\frac{64\mu^2\varepsilon^2}{L_{\min}}}, 240\mu\varepsilon, 120\varepsilon\gamma_u L_f,\\ &\frac{\sqrt{2\gamma_u L_f}\|x^* - x_0\|}{\sqrt{\mu}} + 16\varepsilon\gamma_u L_f + \sqrt{\frac{32\varepsilon^2\mu\gamma_u L_f}{L_{\min}}}\Bigg\}.\end{aligned} \tag{1.18}$$

Proof. We prove the result by induction. Notice $y_0 = v_0 = x_0$. Hence, (1.17) holds for $k = 0$. Suppose that for some $k \geq 0$, it holds

$$\|y_t - x^*\| \leq R,\ \|v_t - x^*\| \leq R,\ \|x_t - x^*\| \leq R,\ \forall\, t = 0, \ldots, k. \tag{1.19}$$

In the following, we show (1.17) also holds for $k+1$.

First, by the μ-strong convexity of ϕ, it holds $\frac{\mu}{2}\|x_{k+1} - x^*\|^2 \leq \phi(x_{k+1}) - \phi(x^*)$. Hence, from (1.13), (1.15), and (1.16), it follows

$$\begin{aligned}
\|x_{k+1} - x^*\|^2 &\leq \frac{1}{\mu A_{k+1}}\left[\|x^* - x_0\|^2 + 2B_{k+1}(x^*) + 2\delta_{k+1}\right] \\
&\leq \frac{1}{\mu A_{k+1}}\|x^* - x_0\|^2 + \frac{2\varepsilon}{A_{k+1}}\sum_{t=0}^{k}\frac{1}{(t+1)^2}\|x^* - x_{t+1}\| \\
&\quad + \frac{2\varepsilon}{A_{k+1}}\sum_{t=0}^{k}\left(\frac{1}{(t+1)^2}\|x_t - x_{t+1}\| + \frac{6}{(t+1)^2}\|y_t - x_{t+1}\| + \frac{4\mu\varepsilon}{L_{\min}(t+1)^4}\right) \\
&\leq \frac{1}{\mu A_{k+1}}\|x^* - x_0\|^2 + \frac{2\varepsilon}{A_{k+1}}\left(2R + \frac{1}{(k+1)^2}\|x^* - x_{k+1}\|\right) \\
&\quad + \frac{2\varepsilon}{A_{k+1}}\left(4R + \frac{1}{(k+1)^2}\|x^* - x_{k+1}\| + 24R + \frac{6}{(k+1)^2}\|x^* - x_{k+1}\| + \frac{8\mu\varepsilon}{L_{\min}}\right) \\
&= \frac{16\varepsilon}{A_{k+1}(k+1)^2}\|x^* - x_{k+1}\| + \frac{1}{\mu A_{k+1}}\|x^* - x_0\|^2 + \frac{2\varepsilon}{A_{k+1}}\left(30R + \frac{8\mu\varepsilon}{L_{\min}}\right),
\end{aligned}$$

where in the third inequality, we have used

$$\sum_{t=0}^{\infty}\frac{1}{(t+1)^4} \leq \sum_{t=0}^{\infty}\frac{1}{(t+1)^2} \leq 2. \tag{1.20}$$

Thus, we obtain

$$\begin{aligned}
\|x_{k+1} - x^*\| &\leq \sqrt{\frac{1}{\mu A_{k+1}}\|x^* - x_0\|^2 + \frac{2\varepsilon}{A_{k+1}}\left(30R + \frac{8\mu\varepsilon}{L_{\min}}\right) + \left(\frac{8\varepsilon}{A_{k+1}(k+1)^2}\right)^2} + \frac{8\varepsilon}{A_{k+1}(k+1)^2} \\
&\leq \frac{\|x^* - x_0\|}{\sqrt{\mu A_{k+1}}} + \frac{16\varepsilon}{A_{k+1}} + \sqrt{\frac{16\mu\varepsilon^2}{A_{k+1}L_{\min}}} + \sqrt{\frac{60\varepsilon R}{A_{k+1}}} \\
&\leq \frac{\sqrt{\gamma_u L_f}\|x^* - x_0\|}{\sqrt{2\mu}} + 8\varepsilon\gamma_u L_f + \sqrt{\frac{8\varepsilon^2\mu\gamma_u L_f}{L_{\min}}} + \sqrt{30\varepsilon\gamma_u L_f R} \\
&\leq R,
\end{aligned} \tag{1.21}$$

where the third inequality holds because $A_{k+1} \geq \frac{2}{\gamma_u L_f}, \forall\, k \geq 0$, and the last one follows from the choice of R.

Second, plugging (1.15) and (1.16) into (1.12) gives

$$\begin{aligned}
\frac{1 + \mu A_{k+1}}{2}\|x^* - v_{k+1}\|^2 &\leq \frac{1}{2}\|x^* - x_0\|^2 + \mu\varepsilon\sum_{t=0}^{k}\frac{1}{(t+1)^2}\|x^* - x_{t+1}\| \\
&\quad + \mu\varepsilon\sum_{t=0}^{k}\left(\frac{1}{(t+1)^2}\|x_t - x_{t+1}\| + \frac{6}{(t+1)^2}\|y_t - x_{t+1}\| + \frac{4\mu\varepsilon}{L_{\min}(t+1)^4}\right) \\
&\leq \frac{1}{2}\|x^* - x_0\|^2 + \mu\varepsilon\left(2R + 4R + 24R + \frac{8\mu\varepsilon}{L_{\min}}\right),
\end{aligned} \tag{1.22}$$

where the second inequality follows from (1.19), (1.20), (1.21), and the triangle inequality. Hence, by $1 + \mu A_{k+1} > 1$, it follows

$$\|x^* - v_{k+1}\| \le \|x^* - x_0\| + \sqrt{\frac{16\mu^2\varepsilon^2}{L_{\min}}} + \sqrt{60\mu\varepsilon R} \le R,$$

where the second inequality holds by the choice of R.

Third, y_{k+1} is a convex combination of x_{k+1} and v_{k+1}, and thus $\|x^* - y_{k+1}\| \le R$. This finishes the induction and completes the proof. □

By Lemma 5, we have from (1.15) and (1.20) that

$$B_k(x^*) \le 2\mu\varepsilon R, \ \forall\, k \ge 0, \tag{1.23}$$

and from (1.16) that

$$\delta_k \le 28\mu\varepsilon R + \frac{8\mu^2\varepsilon^2}{L_{\min}}, \ \forall\, k \ge 0. \tag{1.24}$$

Therefore, plugging (1.23) and (1.24) into (1.13) gives

$$\phi(x_k) - \phi(x^*) \le \frac{E_0}{A_k} := \frac{1}{A_k}\left[\frac{1}{2}\|x^* - x_0\|^2 + 30\mu\varepsilon R + \frac{8\mu^2\varepsilon^2}{L_{\min}}\right], \ \forall\, k \ge 1. \tag{1.25}$$

1.3.2 Iteration Complexity

We have from (1.25) with the μ-strong convexity of ϕ and from (1.22) that

$$\|x_k - x^*\| \le \sqrt{\frac{2E_0}{\mu A_k}}, \qquad \|v_k - x^*\| \le \sqrt{\frac{2E_0}{\mu A_k}}, \qquad \forall\, k \ge 1. \tag{1.26}$$

Since y_k is a convex combination of x_k and v_k, it also holds that

$$\|y_k - x^*\| \le \sqrt{\frac{2E_0}{\mu A_k}}, \qquad \forall\, k \ge 1. \tag{1.27}$$

Hence, by the triangle inequality and L_f-smoothness of f, it follows that

$$\begin{aligned}
&\|\tilde\nabla f(x_{k+1}) - \tilde\nabla f(y_k) - M_k(x_{k+1} - y_k)\| \\
&\le (L_f + M_k)\|x_{k+1} - y_k\| + \|e(x_{k+1})\| + \|e(y_k)\| \\
&\le (L_f + M_k)\|x_{k+1} - y_k\| + 2\tau_k \\
&\le (L_f + \gamma_u L_f)\left(\sqrt{\frac{2E_0}{\mu A_{k+1}}} + \sqrt{\frac{2E_0}{\mu A_k}}\right) + 2\tau_k,
\end{aligned} \tag{1.28}$$

where the last inequality holds by (1.26) and (1.27) and $M_k \le \gamma_u L_f$. Therefore, Algorithm 1 will return x_{k+1} as the output if

$$2(L_f + \gamma_u L_f)\sqrt{\frac{2E_0}{\mu A_k}} + 2\tau_k \le \frac{\varepsilon}{2}. \tag{1.29}$$

On the other side, from the update of x_{k+1}, it holds

$$0 \in \tilde\nabla f(y_k) + M_k(x_{k+1} - y_k) + \partial\Psi(x_{k+1}).$$

Hence, $\nabla f(x_{k+1}) - \tilde\nabla f(y_k) - M_k(x_{k+1} - y_k) \in \partial\phi(x_{k+1})$, and thus

$$\begin{aligned}
\mathrm{dist}\big(0, \partial\phi(x_{k+1})\big) &\le \|\nabla f(x_{k+1}) - \tilde\nabla f(y_k) - M_k(x_{k+1} - y_k)\| \\
&\le \|\tilde\nabla f(x_{k+1}) - \tilde\nabla f(y_k) - M_k(x_{k+1} - y_k)\| + \|e(x_{k+1})\|
\end{aligned}$$

$$\leq \|\tilde{\nabla} f(x_{k+1}) - \tilde{\nabla} f(y_k) - M_k(x_{k+1} - y_k)\| + \tau_k. \tag{1.30}$$

Therefore, when Algorithm 1 stops, x_{k+1} is an ε-stationary solution of ϕ, since $\tau_k \leq \frac{\varepsilon}{2}$ from (1.4).

Summarizing the above analysis, we give the iteration complexity result of Algorithm 1 to produce an ε-stationary solution as follows.

Theorem 2 (Iteration Complexity)**.** Given $\varepsilon > 0$, Algorithm 1 will produce an ε-stationary solution of ϕ within $K_\varepsilon + 1$ iterations, where K_ε satisfies

$$K_\varepsilon \leq \left\lceil \frac{\log 8\sqrt{\frac{\gamma_u L_f E_0}{\mu}} \frac{(1+\gamma_u)L_f}{\varepsilon}}{\log\left(1 + \sqrt{\frac{\mu}{2\gamma_u L_f}}\right)} \right\rceil + 1. \tag{1.31}$$

In particular, if exact gradients of f are used in the algorithm, the total number of iterations $K_\varepsilon + 1$ satisfies

$$K_\varepsilon \leq \left\lceil \frac{\log 8\sqrt{\frac{\gamma_u L_f \|x^* - x_0\|^2}{2\mu}} \frac{(1+\gamma_u)L_f}{\varepsilon}}{\log\left(1 + \sqrt{\frac{\mu}{2\gamma_u L_f}}\right)} \right\rceil + 1. \tag{1.32}$$

Proof. Notice that $k+1$ iterations are needed to output x_{k+1}. Hence, it suffices to solve the inequality in (1.29) for k. When k is the upper bound in (1.31), it holds that

$$\left(1 + \sqrt{\frac{\mu}{2\gamma_u L_f}}\right)^{k-1} \geq 8\sqrt{\frac{\gamma_u L_f E_0}{\mu}} \frac{(1+\gamma_u)L_f}{\varepsilon}.$$

Hence, it follows from (1.14) that

$$\sqrt{A_k} \geq 8\sqrt{\frac{2}{\gamma_u L_f}} \sqrt{\frac{\gamma_u L_f E_0}{\mu}} \frac{(1+\gamma_u)L_f}{\varepsilon}, \tag{1.33}$$

and thus

$$2(1+\gamma_u)L_f\sqrt{\frac{2E_0}{\mu A_k}} \leq \frac{\varepsilon}{4}. \tag{1.34}$$

Now adding $2\tau_k \leq \frac{\varepsilon}{4}$ and (1.34) gives (1.29).

Finally, notice that when exact gradients of f are used, it holds $B_k(\cdot) \equiv 0$ and $\delta_k = 0$ for all k. Then $\phi(x_k) - \phi(x^*) \leq \frac{1}{2A_k}\|x^* - x_0\|^2$, i.e., (1.25) holds with $E_0 = \frac{1}{2}\|x^* - x_0\|^2$. Thus we have (1.32) from (1.31) and complete the proof. □

Remark 2. Because each while-loop must exit in at most $\left\lceil \log_{\gamma_u} \frac{L_f}{L_{\min}} \right\rceil + 1$ steps and each step takes two gradients, the total number of inexact gradient evaluations will be $2K_\varepsilon \left\lceil \log_{\gamma_u} \frac{L_f}{L_{\min}} \right\rceil + 2K_\varepsilon$. When $\sqrt{\frac{\mu}{2\gamma_u L_f}} \leq 1$, it holds $\log\left(1 + \sqrt{\frac{\mu}{2\gamma_u L_f}}\right) \geq \sqrt{\frac{\mu}{8\gamma_u L_f}}$. Otherwise, by monotonicity of $\log(\cdot)$, it holds $\log\left(1 + \sqrt{\frac{\mu}{2\gamma_u L_f}}\right) \geq \log 2$. Hence, we have

$$K_\varepsilon \leq \left\lceil \left(\sqrt{\frac{8\gamma_u L_f}{\mu}} + \frac{1}{\log 2}\right) \log 8\sqrt{\frac{\gamma_u L_f E_0}{\mu}} \frac{(1+\gamma_u)L_f}{\varepsilon} \right\rceil + 1.$$

Thus the complexity of (inexact) gradient evaluations is upper bounded by

$$2\left(\left\lceil\left(\sqrt{\frac{8\gamma_u L_f}{\mu}}+\frac{1}{\log 2}\right)\log 8\sqrt{\frac{\gamma_u L_f E_0}{\mu}}\frac{(1+\gamma_u)L_f}{\varepsilon}\right\rceil+1\right)\left(\left\lceil\log_{\gamma_u}\frac{L_f}{L_{\min}}\right\rceil+1\right).$$

Proposition 1. Given $\varepsilon>0$, let K_ε be the iteration number, at which Algorithm 1 produces an ε-stationary solution of ϕ. Then

$$A_k \le \left(1+\frac{2\mu}{L_{\min}}+2\sqrt{\frac{2\mu}{L_{\min}}}\right)^2\left(\frac{128E_0}{\mu}\frac{(1+\gamma_u)^2L_f^2}{\varepsilon^2}+\frac{2+\sqrt{\frac{L_{\min}}{8\mu}}}{2\mu+2\sqrt{2\mu L_{\min}}}\right), \forall\, k\le K_\varepsilon+1. \tag{1.35}$$

Proof. From the proof of Theorem 2, we see that if (1.33) holds, then Algorithm 1 will return x_{k+1} that is an ε-stationary solution of ϕ. Hence, before the algorithm stops, i.e., reaching to K_ε iterations, (1.33) cannot hold. Therefore, it must hold

$$A_k < \left(8\sqrt{\frac{2E_0}{\mu}}\frac{(1+\gamma_u)L_f}{\varepsilon}\right)^2=\frac{128E_0}{\mu}\frac{(1+\gamma_u)^2L_f^2}{\varepsilon^2}, \forall\, k<K_\varepsilon. \tag{1.36}$$

In addition, by the update rule of A_{k+1} and a_{k+1}, we have that for $k\ge 0$,

$$\frac{(A_{k+1}-A_k)^2}{A_{k+1}}=\frac{2(1+\mu A_k)}{M_k}.$$

Solving the above equation for A_{k+1} gives

$$\begin{aligned}
A_{k+1} &= A_k+\frac{1+\mu A_k}{M_k}+\sqrt{\left(A_k+\frac{1+\mu A_k}{M_k}\right)^2-A_k^2}\\
&= A_k+\frac{1+\mu A_k}{M_k}+\sqrt{\left(\frac{1+\mu A_k}{M_k}\right)^2+2A_k\frac{1+\mu A_k}{M_k}}\\
&\le A_k+\frac{2(1+\mu A_k)}{M_k}+\sqrt{\frac{2A_k}{M_k}}+\sqrt{\frac{2\mu}{M_k}}A_k\\
&\le A_k+\frac{2(1+\mu A_k)}{M_k}+2\sqrt{\frac{2\mu}{M_k}}A_k+\sqrt{\frac{1}{8\mu M_k}}\\
&= \left(1+\frac{2\mu}{M_k}+2\sqrt{\frac{2\mu}{M_k}}\right)A_k+\frac{2}{M_k}+\sqrt{\frac{1}{8\mu M_k}}\\
&\le \left(1+\frac{2\mu}{L_{\min}}+2\sqrt{\frac{2\mu}{L_{\min}}}\right)A_k+\frac{2}{L_{\min}}+\sqrt{\frac{1}{8\mu L_{\min}}},
\end{aligned}$$

where the last inequality holds because $M_k\ge L_{\min}$. The above inequality together with (1.36) implies that for $k=K_\varepsilon, K_\varepsilon+1$,

$$A_k \le \left(1+\frac{2\mu}{L_{\min}}+2\sqrt{\frac{2\mu}{L_{\min}}}\right)^2\left(\frac{128E_0}{\mu}\frac{(1+\gamma_u)^2L_f^2}{\varepsilon^2}+\frac{\frac{2}{L_{\min}}+\sqrt{\frac{1}{8\mu L_{\min}}}}{\frac{2\mu}{L_{\min}}+2\sqrt{\frac{2\mu}{L_{\min}}}}\right).$$

This completes the proof. □

Recommended Resources

Articles

[1] Yu Nesterov. “Gradient methods for minimizing composite functions”. In: *Mathematical programming* 140.1 (2013), pp. 125–161. (pp. 1, 3–5)

[2] Mark Schmidt, Nicolas Roux, and Francis Bach. “Convergence rates of inexact proximal-gradient methods for convex optimization”. In: *Advances in neural information processing systems* 24 (2011). (p. 1)

[3] Qihang Lin and Yangyang Xu. “Reducing the complexity of two classes of optimization problems by inexact accelerated proximal gradient method”. In: *SIAM Journal on Optimization* 33.1 (2023), pp. 1–35. (p. 1)